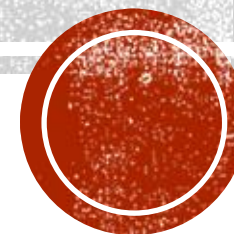


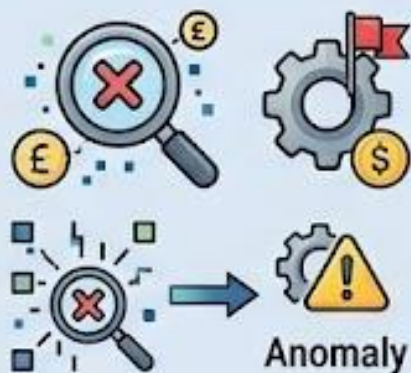
VARIATIONAL AUTOENCODERS

Laxmi Narayan



FIVE KEY USE CASES OF VARIATIONAL AUTOENCODERS (VAEs)

1. ANOMALY DETECTION (FINANCE & MANUFACTURING)



Detecting Fraudulent Transactions

Quality Control in Assembly Lines

Scanning for Defects in Images

2. HEALTHCARE & DRUG DISCOVERY



Generating New Molecular Compounds

Creating Synthetic Patient Data (Privacy)

Exploring Chemical Space

3. RECOMMENDATION SYSTEMS (COLLABORATIVE FILTERING)



Understanding Complex User Preferences

Predicting Movie/Product Ratings

Filling Missing Data Points

4. IMAGE PROCESSING & COMPUTER VISION



Image Denoising & Restoring Quality

Image Completion (Hallucinating Parts)

Manipulating Specific Attributes

5. GAME DESIGN & CREATIVE ARTS

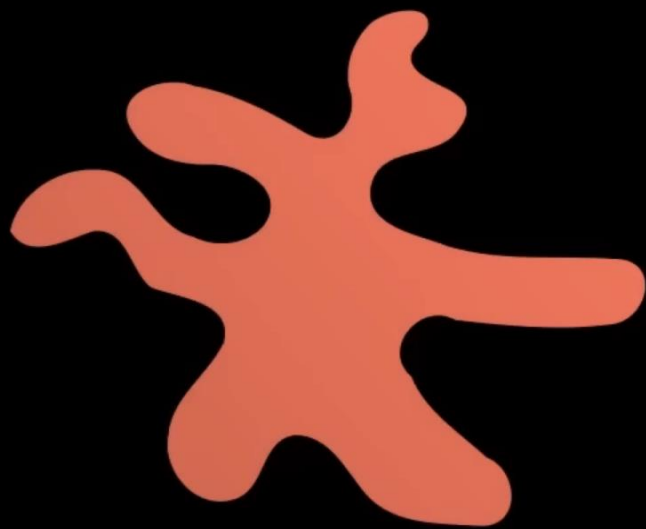


Procedural Level Generation

Dynamic Content Creation

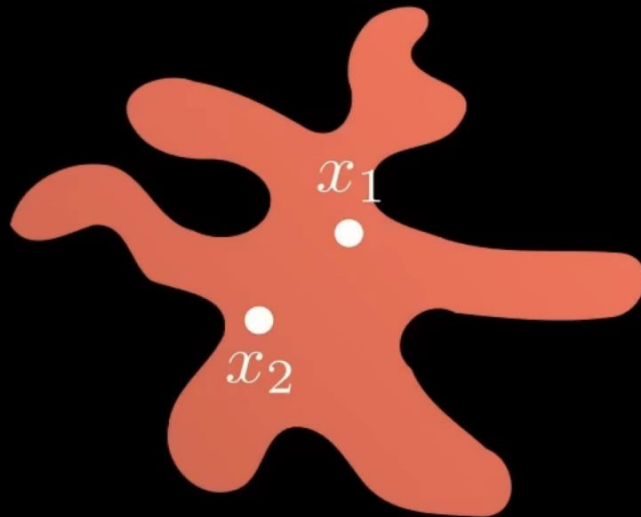
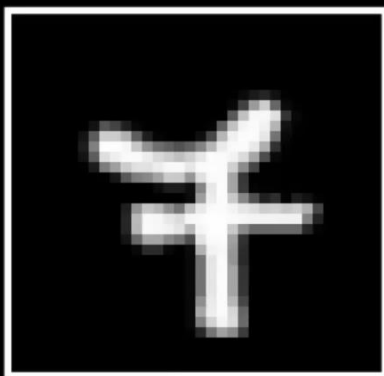
Style Transfer & Trait Editing

$$p(x)$$



Data Distribution

$$p(x)$$



Data Distribution

$$p(x)$$



Data Distribution

$$p(z)$$



Latent Distribution

$$p(x)$$



Data Distribution

$$p(z)$$



Latent Distribution

$$p(x)$$



Data Distribution

$$p(z|x)$$



$$p(z)$$



Latent Distribution

$$p(x)$$



Data Distribution

$$p(z|x)$$



$$p(x|z)$$

$$p(z)$$



Latent Distribution



$$p(x)$$



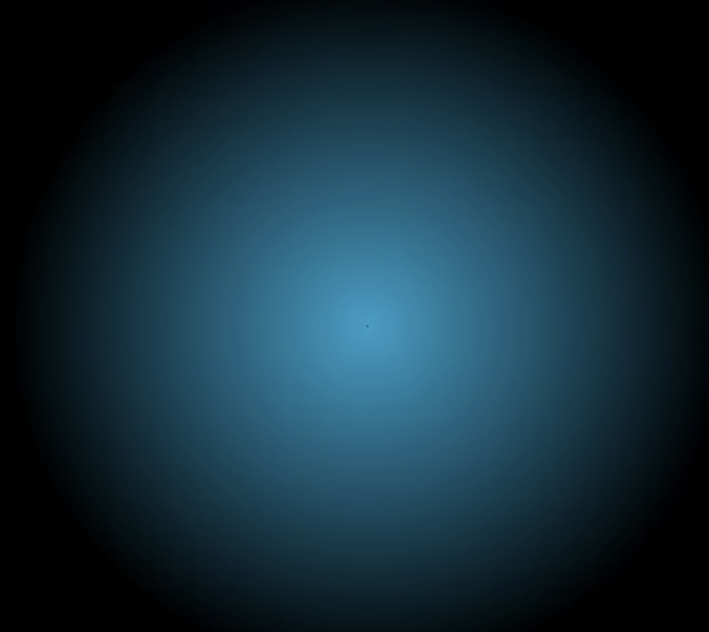
Data Distribution

$$p(z|x)$$



$$p(x|z)$$

$$p(z) = \mathcal{N}(0, 1)$$



Latent Distribution

$$p(x)$$



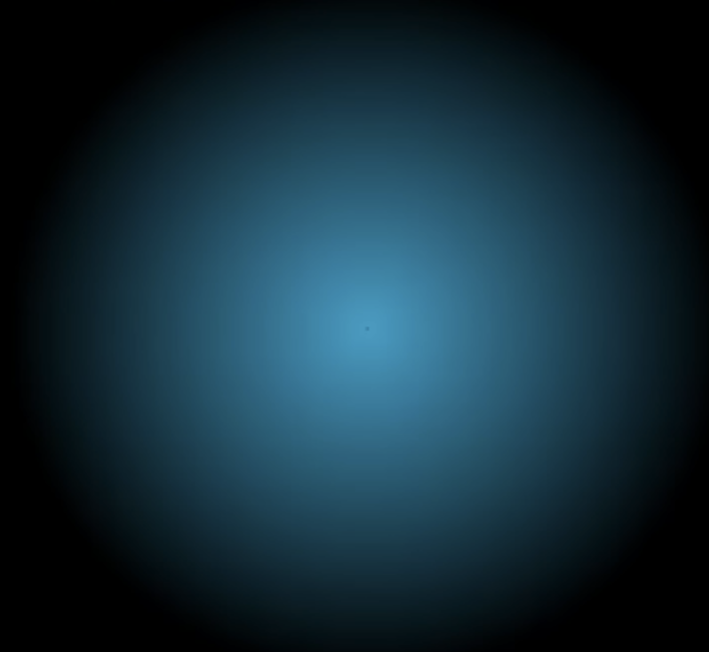
Data Distribution

$$q(z|x) \approx p(z|x)$$



$$p(x|z)$$

$$p(z) = \mathcal{N}(0, 1)$$



Latent Distribution

Variational Bayes

$$p(x)$$



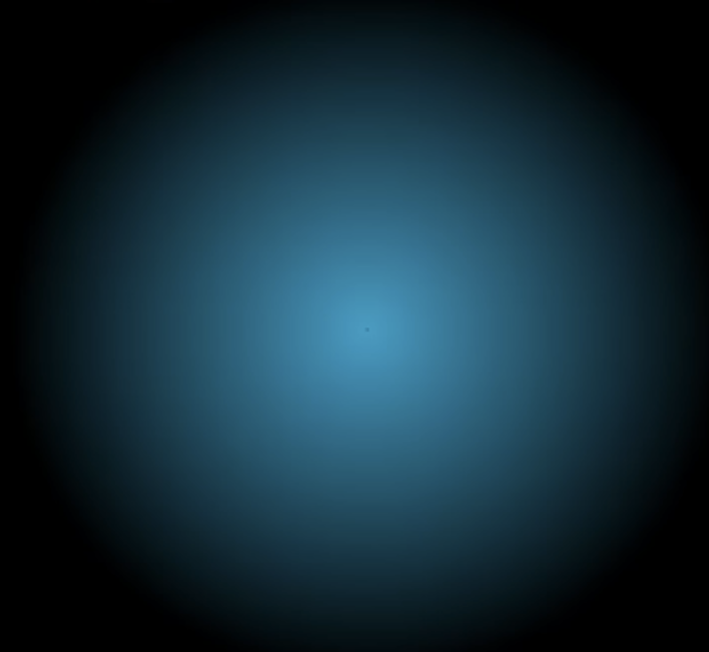
Data Distribution

$$q(z|x) = \mathcal{N}(\mu, \sigma)$$



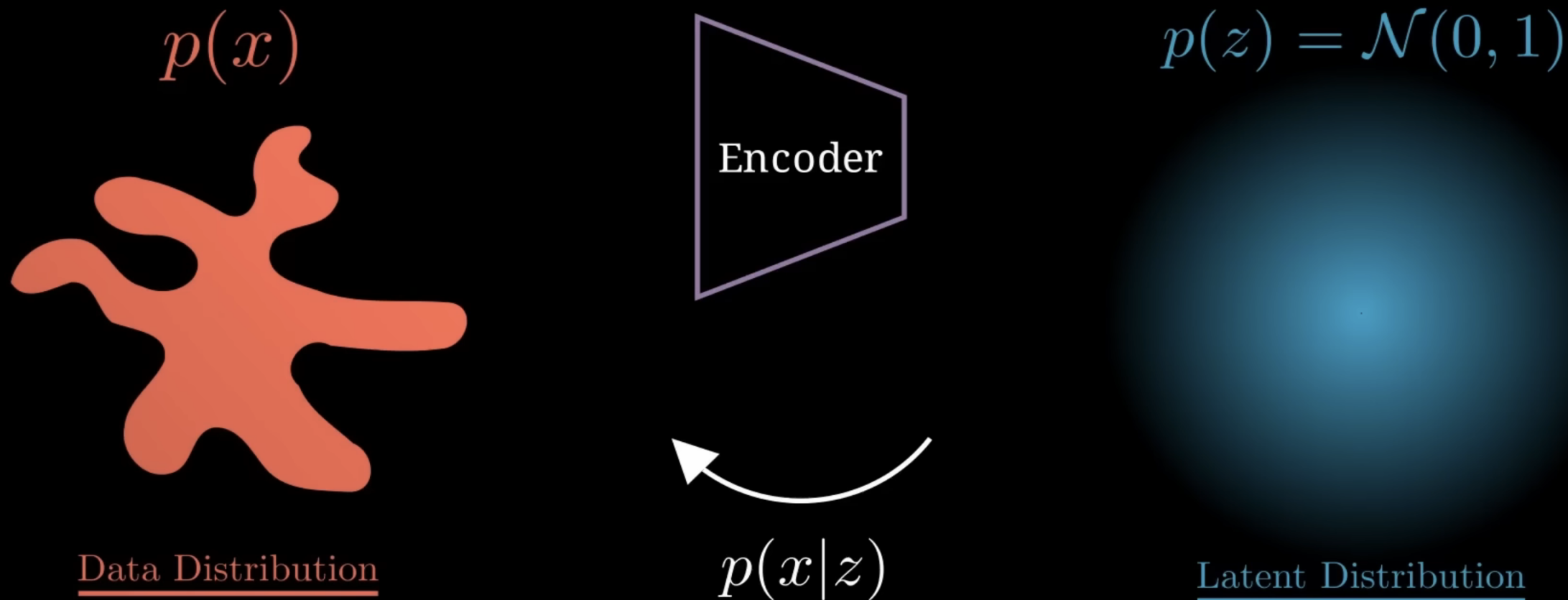
$$p(x|z)$$

$$p(z) = \mathcal{N}(0, 1)$$

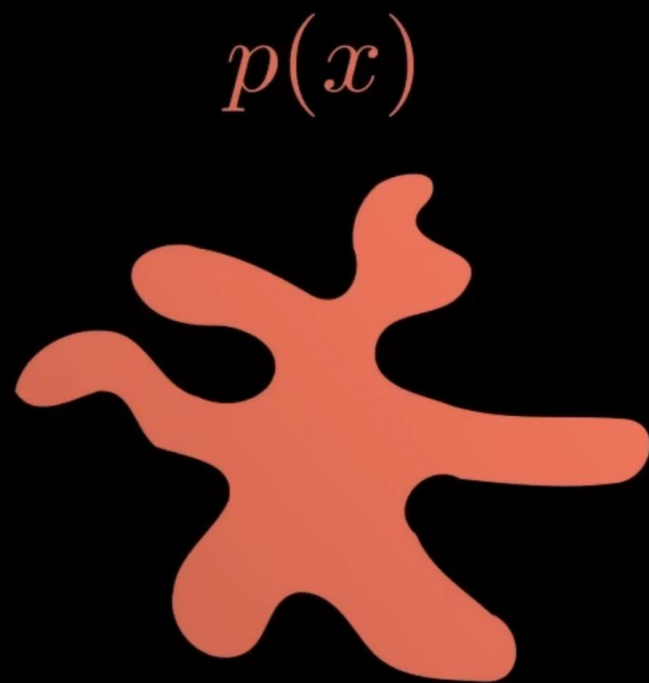


Latent Distribution

Autoencoder Variational Bayes



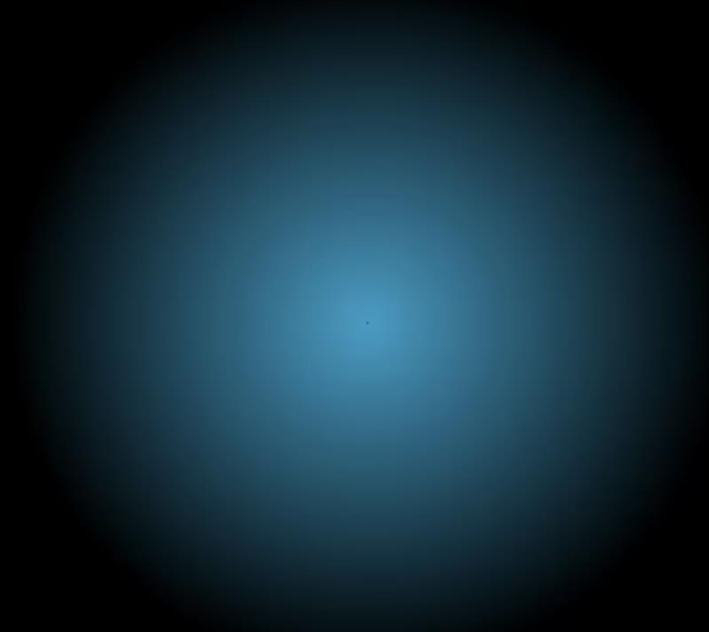
Autoencoder Variational Bayes



Data Distribution

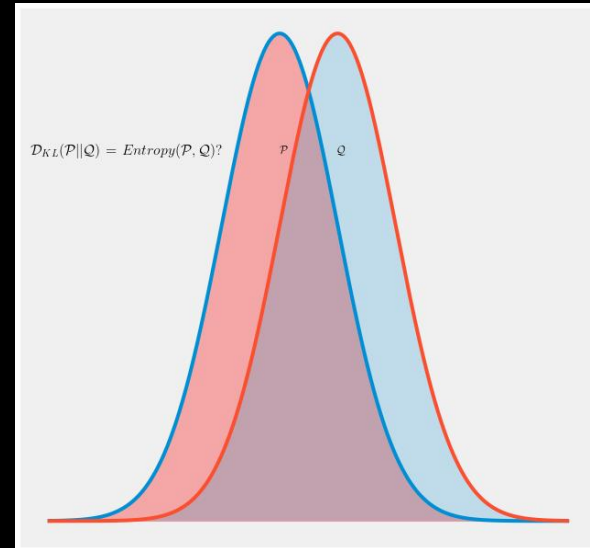


$$p(z) = \mathcal{N}(0, 1)$$



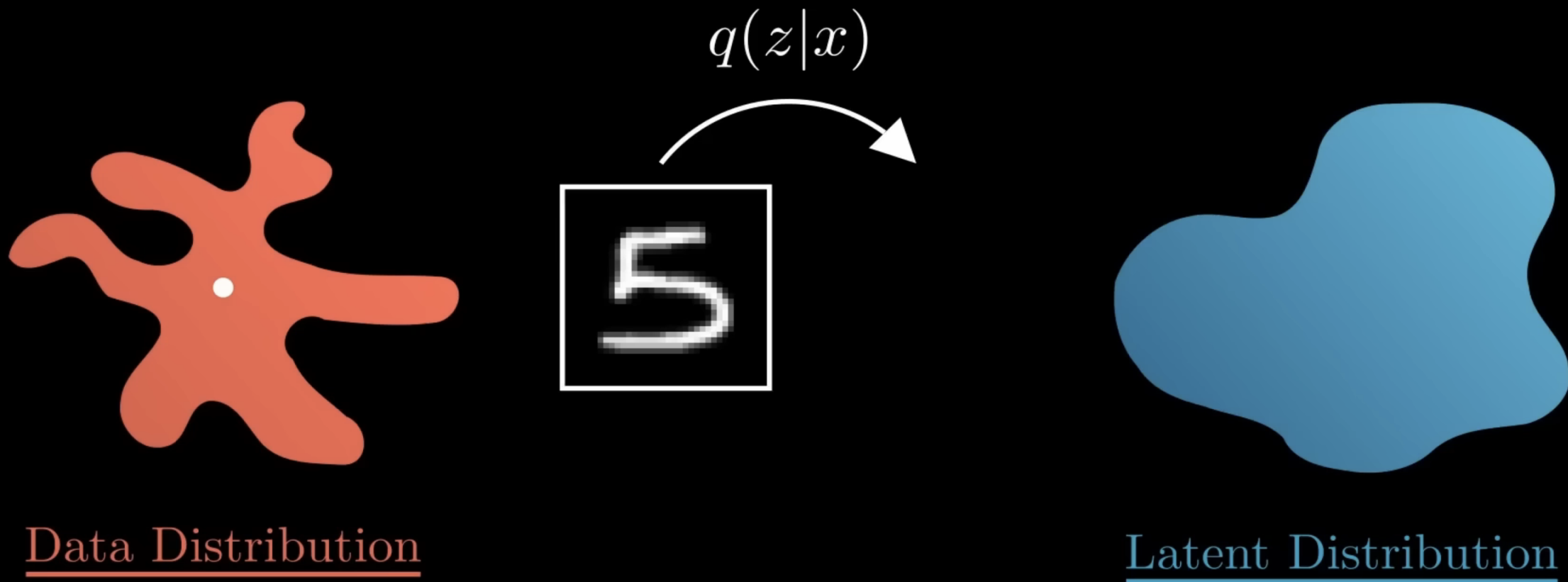
Latent Distribution

$$\mathcal{L}(x) = \mathbb{E}_{q(z|x)} [\log p(x|z)] - \text{KL}(q(z|x) \mid p(z))$$

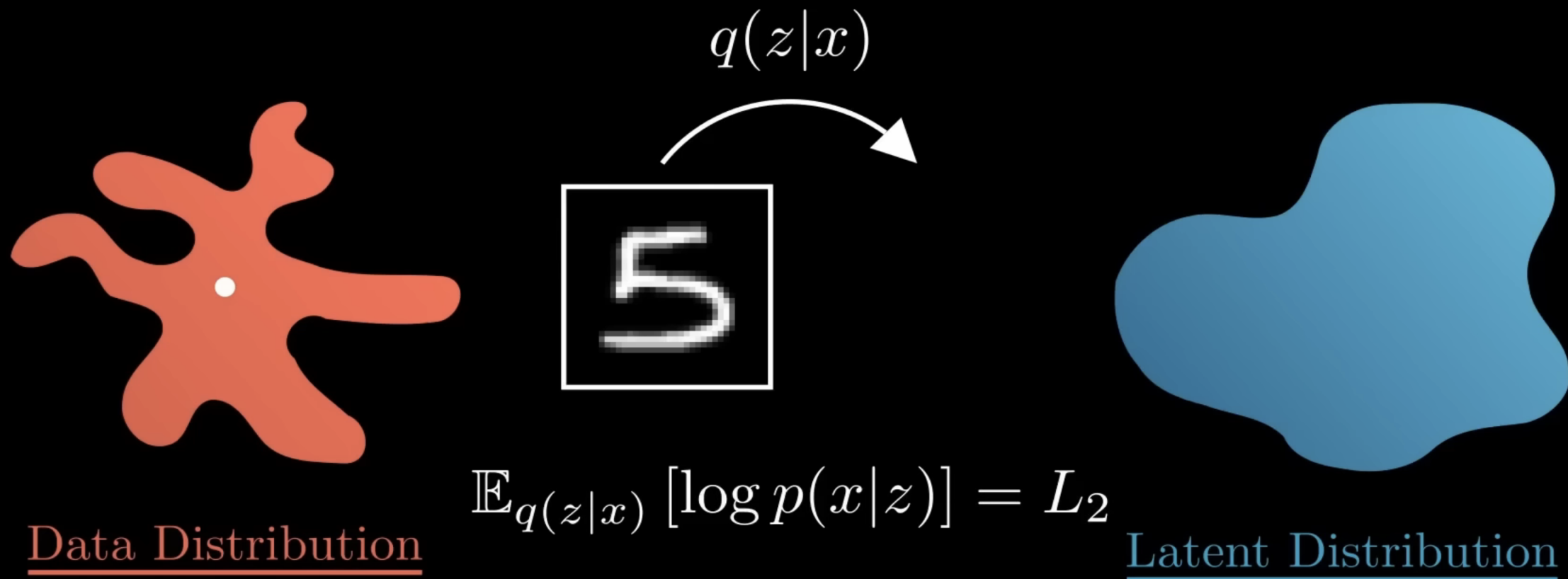


$$\mathcal{L}(x) = \underbrace{\mathbb{E}_{q(z|x)} [\log p(x|z)]}_{\text{Data consistency}} - \overbrace{\text{KL}(q(z|x) \mid p(z))}^{\text{Regularization}}$$

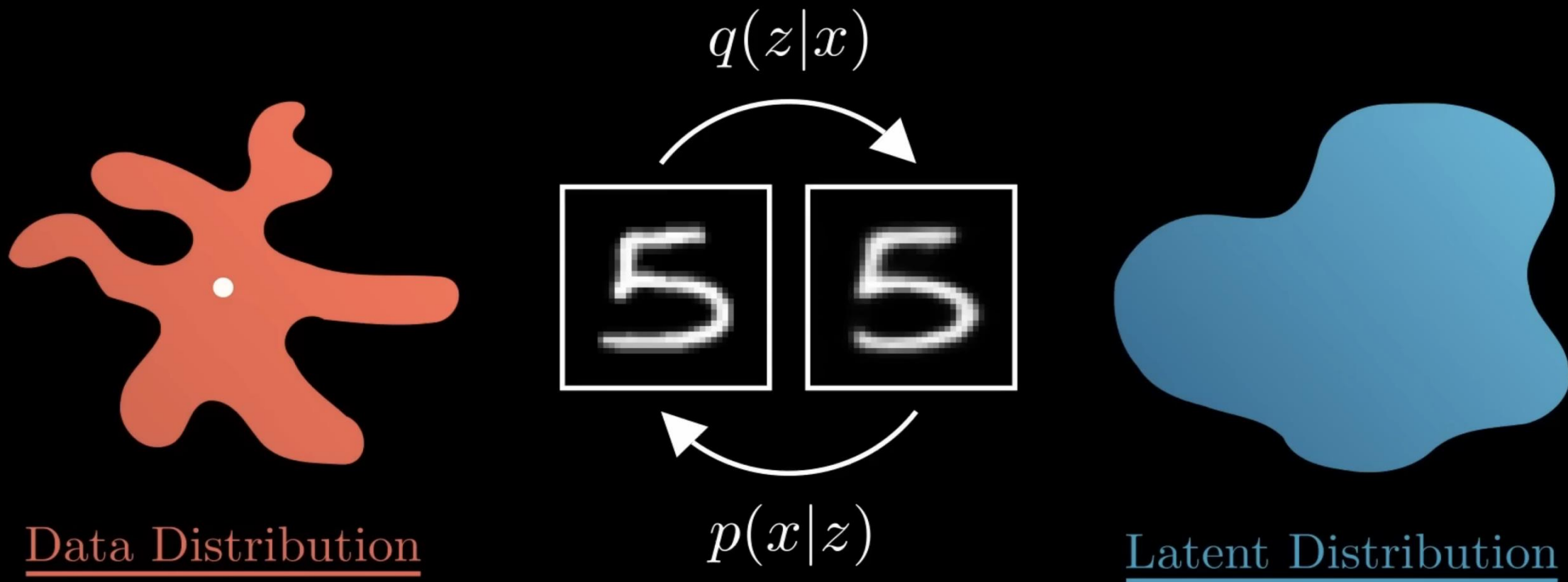
$$\mathcal{L}(x) = \mathbb{E}_{q(z|x)} [\log p(x|z)] - \text{KL}(q(z|x) \parallel p(z))$$



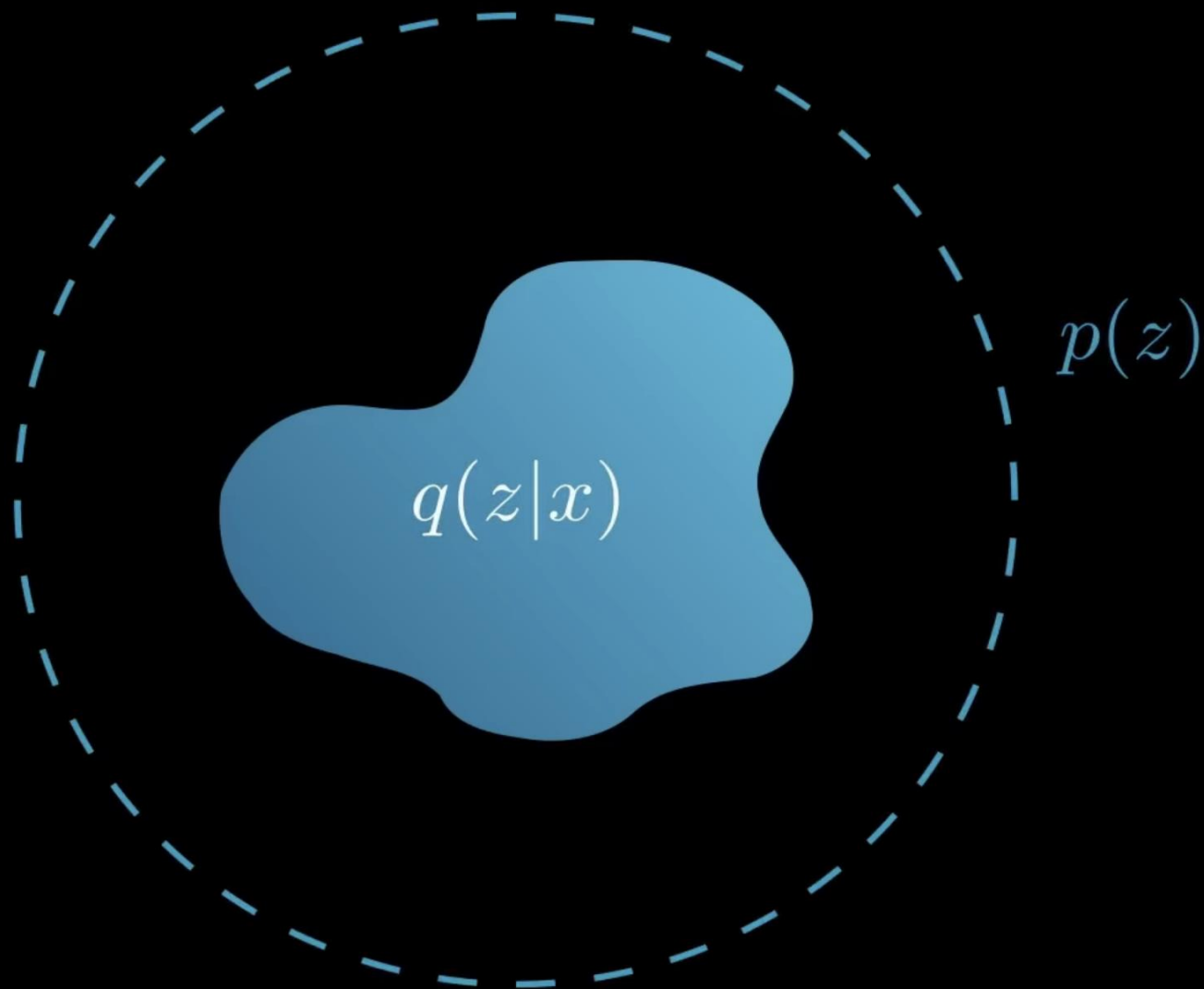
$$\mathcal{L}(x) = \mathbb{E}_{q(z|x)} [\log p(x|z)] - \text{KL}(q(z|x) \mid p(z))$$



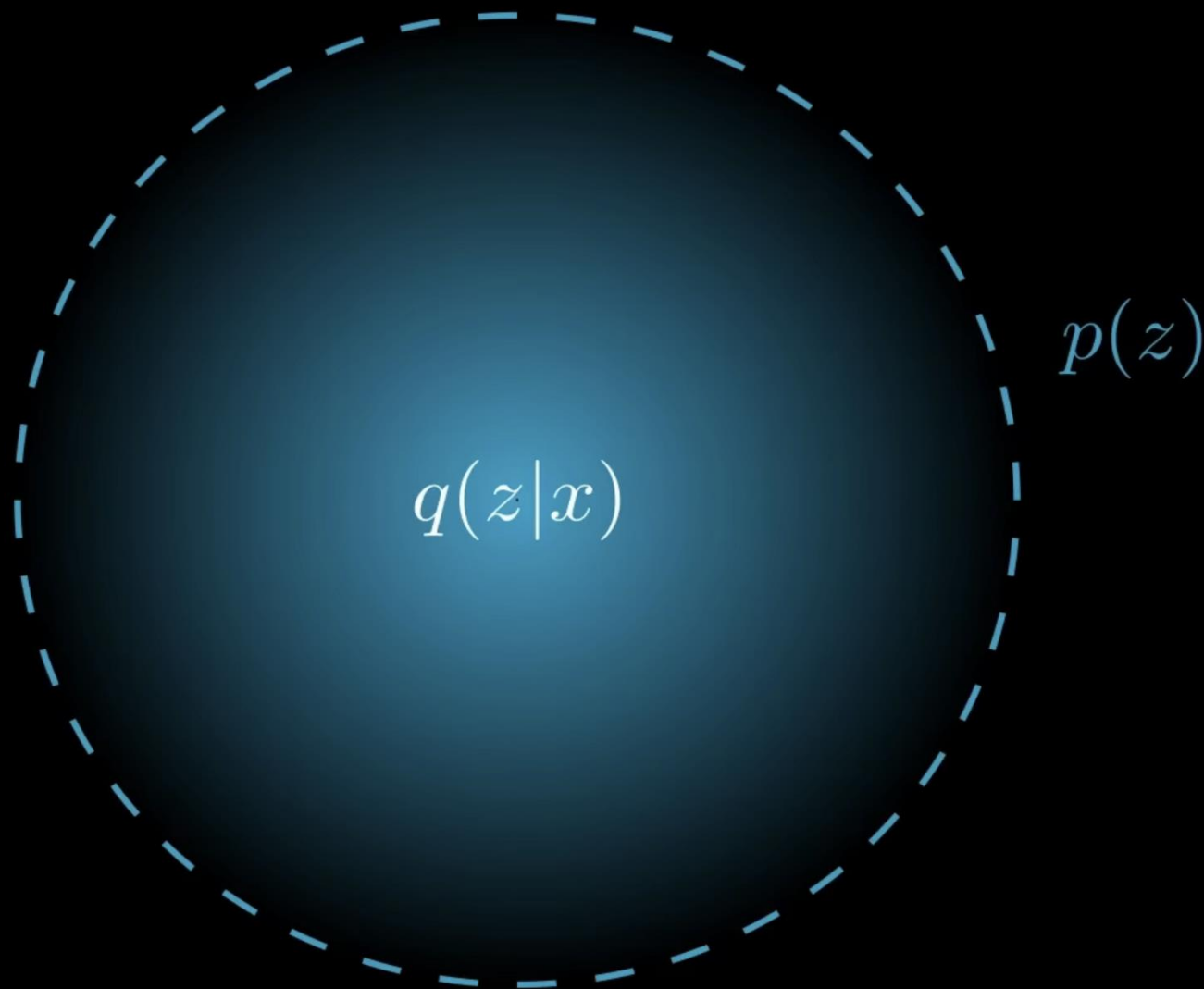
$$\mathcal{L}(x) = \mathbb{E}_{q(z|x)} [\log p(x|z)] - \text{KL}(q(z|x) \parallel p(z))$$



$$\mathcal{L}(x) = \mathbb{E}_{q(z|x)} [\log p(x|z)] - \text{KL}(q(z|x) \mid p(z))$$



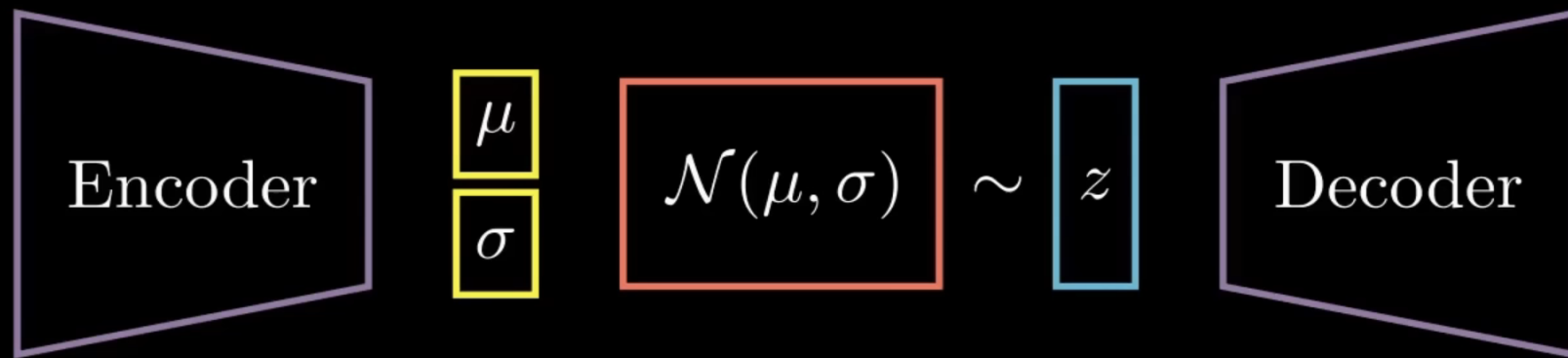
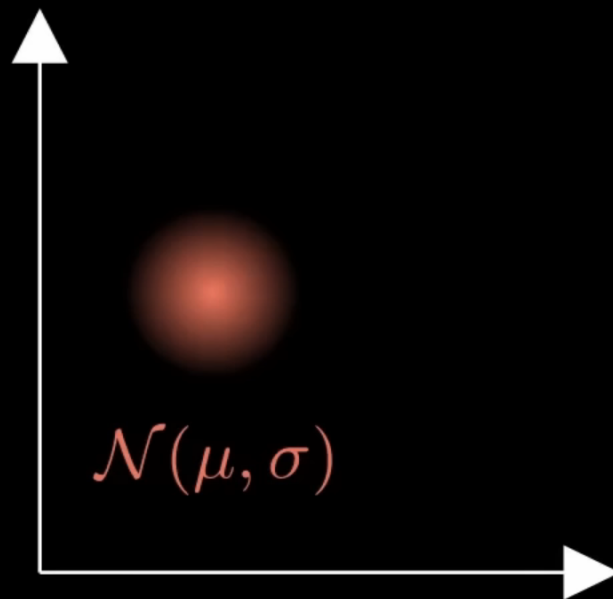
$$\mathcal{L}(x) = \mathbb{E}_{q(z|x)} [\log p(x|z)] - \text{KL}(q(z|x) \mid p(z))$$



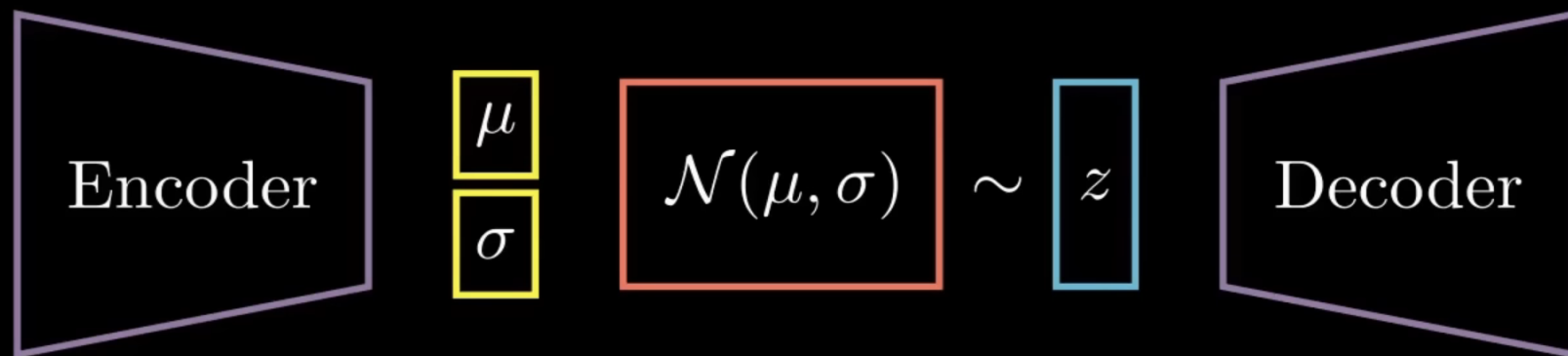
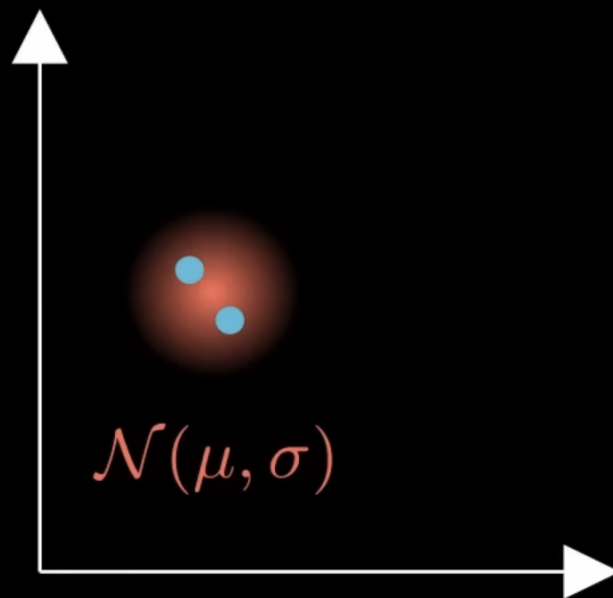
Latent space regularization

$$\mathcal{L}(x) = \underbrace{\mathbb{E}_{q(z|x)} [\log p(x|z)]}_{\text{L2}} - \overbrace{\text{KL}(q(z|x) \mid p(z))}$$

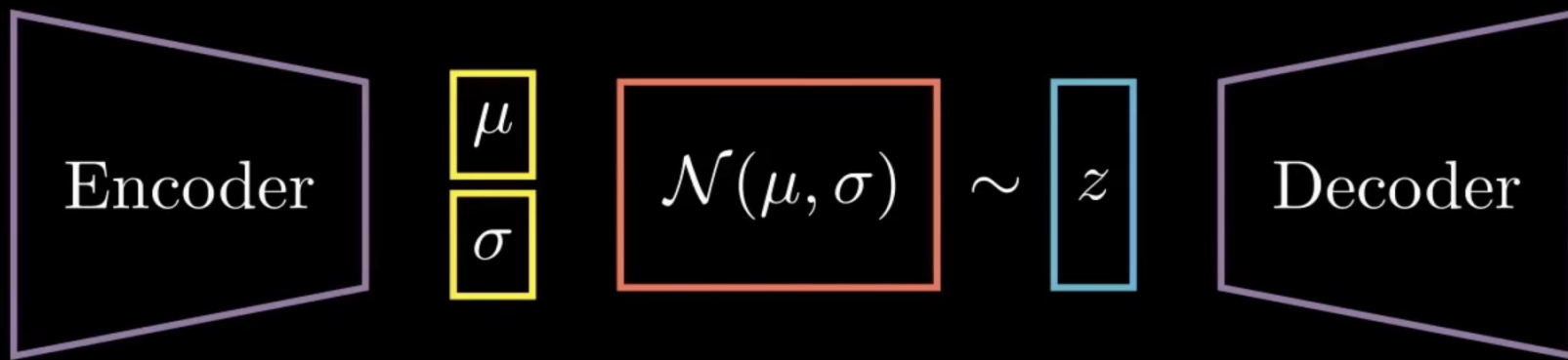
Latent Space



Latent Space



$$\mathcal{L}(x, x') = L_2(x, x') + D_{KL}(\mathcal{N}(\mu, \sigma) \mid \mathcal{N}(0, 1))$$



$$\mathcal{L}(x, x') = L_2(x, x') + D_{KL}(\mathcal{N}(\mu, \sigma) \mid \mathcal{N}(0, 1))$$

